

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 07)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops

☐

B Steam washing

☐

C Inflatable booms

☐

D Polymerising agents

☐

0 1 . 2

Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 1 . 3

Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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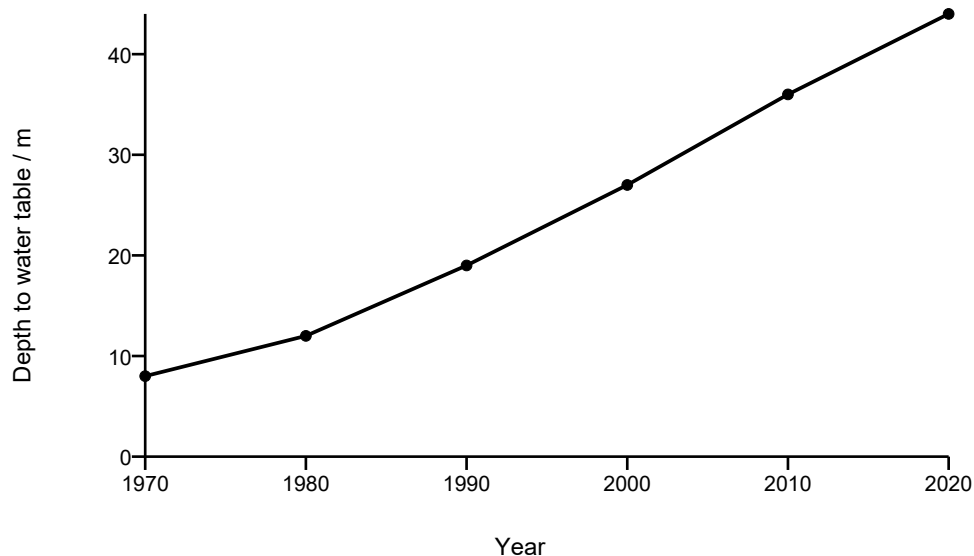
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5

0 2

Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1**0 2****. 1**

Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]**Answer**

m/yr

0 2**. 2**

Explain why the water table is falling.

[2 marks]

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0 2**. 3**

Explain the features a rock must have to form an aquifer.

[3 marks]

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0 2 . 4

Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

1

2

3

4

10

0 3 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content

☐

B High clay content

☐

C Crumb peds

☐

D High pH

☐

0 3 . 2

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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03 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

5

04 . 1

Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

04 . 2

Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

0 4 . 3

Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

1

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2

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3

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0 4 . 4

Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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10

0 5 . 1

Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☒

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else ☒ can be assumed to be safe

D To measure the concentration of a pollutant at the point of release ☐

0 5 . 2

Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 5 . 3

A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

5**0 6 . 1**

Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

0 6 . 2

Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

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Properties causing ozone depletion

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0 6 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects. **[3 marks]**

1

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2

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3

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0 6 . 4 State what is meant by an **ozone hole**. **[1 mark]**

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0 6 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion. **[1 mark]**

06 . 6

Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

15

07 . 1

Explain how energy absorbed in tropical regions is distributed to higher latitudes such as the UK.

[4 marks]

07 . 2

Explain why the UK has milder winters than Moscow or Churchill, Canada, which are at similar latitudes.

[2 marks]

07 . 3

Explain why the atmosphere would not retain liquid water if atmospheric pressure were much lower.

[2 marks]

07 . 4

Give **two** industrially important gases extracted from the atmosphere.

[2 marks]

1

2

10

08 . 1

Describe how soil texture is determined using a stack of sieves.

[3 marks]

08 . 2

Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

08 . 3

Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

08 . 4

Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

10

09

Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	Cadmium in rice / mg kg^{-1}	Cadmium in kidney of residents / mg kg^{-1}	Cases of bone disease (itai-itai) per 1000 people
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

09 . 1

Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

Answer

times

09 . 2

Explain why cadmium and zinc pollution often occur together.

[1 mark]

09 . 3

Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

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09 . 4

Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

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2

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09 . 5

Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

[2 marks]

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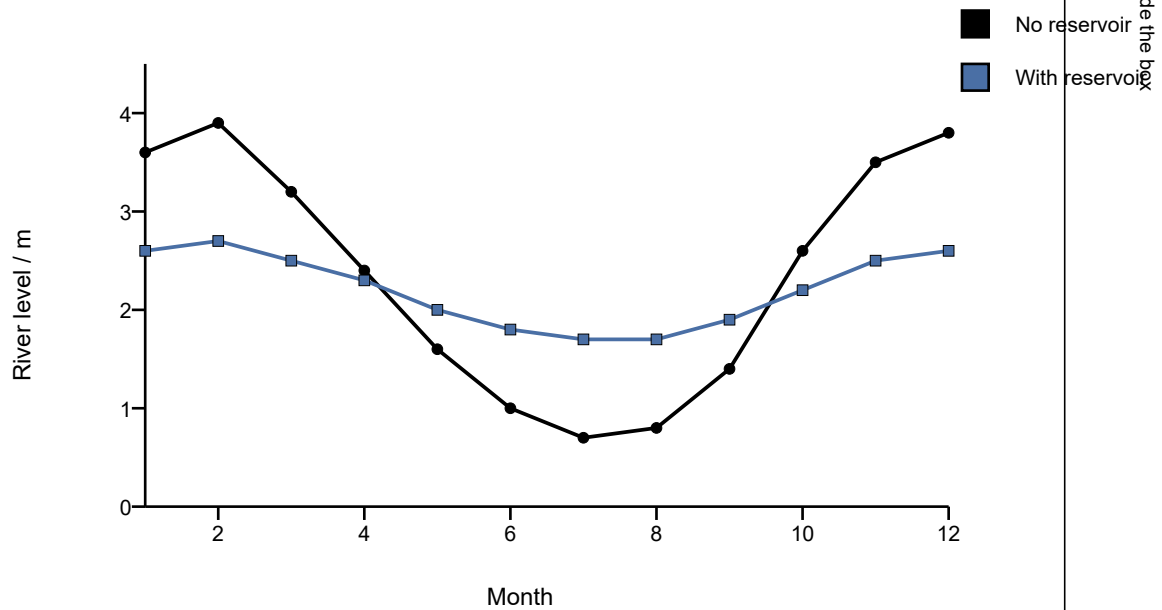
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Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0

1

Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

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1 0

2

Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0

3

Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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10 . 4

Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Handwriting practice area with 25 horizontal dotted lines.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

END OF QUESTIONS